

Advanced Power BI Reference Card

DAX * Time Intelligence * RANKX * Copilot * Advanced Interactivity
Data Analyst Career Programme * Module 5 Advanced * Keep this open while working on your project

Time Intelligence Patterns

```
// YTD - resets at start of each year
Revenue YTD = TOTALYTD([Total Revenue],
DateTable[Date])

// Same period last year
Revenue SPLY =
    CALCULATE([Total Revenue],
        SAMEPERIODLASTYEAR(DateTable[Date]))

// YoY change %
YoY % = DIVIDE([Total Revenue]-[Revenue SPLY],[Revenue
SPLY])

// Rolling 3-month average
Rev Rolling 3M =
VAR vLast = MAX(DateTable[Date])
RETURN CALCULATE(AVERAGE(Sales[revenue_gbp]),
    DATESINPERIOD(DateTable[Date],vLast,-3,MONTH))

// Month-on-Month change
Revenue Prior Month =
    CALCULATE([Total Revenue],
        DATEADD(DateTable[Date],-1,MONTH))

// Running total (cumulative)
Rev Cumulative =
VAR vMax = MAX(DateTable[Date])
RETURN CALCULATE([Total Revenue],
    DateTable[Date] <= vMax, ALL(DateTable))
```

Time intelligence requires:

- * DateTable marked as date table
- * Active relationship: DateTable[Date] to fact[date_col]
- * Filter context must include a time period

RANKX and Filter Context Patterns

```
// Company-wide rank (removes ALL store filters)
Rank (Company) =
VAR v = [Total Revenue]
RETURN RANKX(ALL(Stores),[Total Revenue],v,DESC,DENSE)

// Region rank (keeps region, removes store)
Rank (Region) =
VAR v = [Total Revenue]
RETURN RANKX(
    ALLEXCEPT(Stores, Stores[region]),
    [Total Revenue], v, DESC, DENSE)

// Brand-region rank (keeps both)
Rank (Brand+Region) =
VAR v = [Total Revenue]
RETURN RANKX(
    ALLEXCEPT(Stores, Stores[region], Stores[brand]),
    [Total Revenue], v, DESC, DENSE)

// vs Regional Average
vs Region Avg % =
VAR vRev = [Total Revenue]
VAR vAvg = CALCULATE([Total Revenue],
    ALLEXCEPT(Stores,Stores[region])) /
    CALCULATE(DISTINCTCOUNT(Stores[store_id]),
        ALLEXCEPT(Stores,Stores[region]))
RETURN DIVIDE(vRev - vAvg, vAvg)

// Bottom quartile flag
Is Bottom Quartile =
VAR vRank = [Rank (Region)]
VAR vN = CALCULATE(DISTINCTCOUNT(Stores[store_id]),
    ALLEXCEPT(Stores,Stores[region]))
RETURN vRank >= vN - CEILING(vN*0.25,1) + 1
```

EZ Adjustment: SUMX Pattern

```
// SUMX iterates row-by-row - use when
// different rows need different logic
EZ Adjusted Cost =
SUMX(Sales,
    VAR vCost = Sales[cost_gbp]
    VAR vIsEZ = RELATED(Stores[enterprise_zone])
    VAR vRelief =
        RELATED(Stores[ez_tax_relief_pct])/100
    RETURN IF(vIsEZ, vCost*(1-vRelief), vCost))

EZ Adj Gross Margin % =
VAR vRev = [Total Revenue]
VAR vProfit = vRev - [EZ Adjusted Cost]
RETURN IF(vRev=0,BLANK(),DIVIDE(vProfit,vRev))
```

Advanced Features Checklist

Feature	How to Activate
Field Parameter	Modelling > New Parameter > Fields
What-If Parameter	Modelling > New Parameter > Numeric Range
Drill-Through	Add field to Drill-Through well in Filters pane
Bookmark	View > Bookmarks > Add bookmark
Report Tooltip	Page format > Allow use as tooltip = On
Dynamic Title	Use SELECTEDVALUE in a measure, add to visual title
Conditional Format	Visual > Format > Conditional formatting > Field value

Copilot Evaluation Framework

Rate as	When to Use It
Correct	Right chart type, correct data, valid business insight
Needs work	Roughly right but wrong label, axis or framing - fixable
Wrong	Fabricated numbers, wrong measure, misleading chart type

VAR...RETURN: Every Measure Must Use This

```
// Standard pattern - use this for every measure
Measure Name =
VAR vStep1 = SUM( Sales[revenue_gbp] )
VAR vStep2 = SUM( Sales[cost_gbp] )
VAR vResult = vStep1 - vStep2
RETURN
    IF( vStep1 = 0, BLANK(), vResult )

// Why: variables are calculated once and cached.
// Referencing [Total Revenue] twice in a measure
// evaluates it twice. VAR evaluates it once.

// Debugging: add a RETURN vStep1 temporarily
// to inspect intermediate values.
```

CALCULATE: Key Patterns

```
// Remove all filters on Stores
CALCULATE([Total Revenue], ALL(Stores))

// Remove store filter, keep region filter
CALCULATE([Total Revenue],
    ALLEXCEPT(Stores,Stores[region]))

// Add a filter: only Premium brand
CALCULATE([Total Revenue], Stores[brand]="Premium")

// Activate an inactive relationship
CALCULATE(SUM(Targets[revenue_target_gbp]),
    USERELATIONSHIP(DateTable[Date],Targets[target_month]))

// Combine filters: Premium in South East
CALCULATE([Total Revenue],
    Stores[brand]="Premium",
    Stores[region]="South East")
```

Star Schema for This Project

Table	Type	Joins To
Stores	Dimension (1 side)	Sales, Targets
Categories	Dimension (1 side)	Sales
Sales	Fact (many side)	Stores, Categories, DateTable
Targets	Fact (many side)	Stores, DateTable
DateTable	Date dimension	Sales, Targets

Two fact tables sharing Stores creates ambiguity.
Use USERELATIONSHIP when you need to bring Targets data into a Sales-context measure. Both relationships to Stores must be active. Set cross-filter to Single on both.

Common RANKX Errors

Error	Cause	Fix
All stores show rank 1	ALL() or ALLEXCEPT() not removing filters	Check the first argument to RANKX
Rank shows blank for some stores	Measure returns blank for those stores	Wrap in IF(ISBLANK(...),BLANK()),RANKX(...)
Region rank same as company rank	ALLEXCEPT missing the region column	Add Stores[region] to ALLEXCEPT
SPLY returns same as current	DateTable not marked as date table	Mark DateTable: right-click > Mark as date table

25-Plus Measures Checklist

Foundation (8)

- * Total Revenue, Total Cost, Gross Profit
- * Gross Margin %, Total Discount, Discount Rate %
- * Revenue Target (USERELATIONSHIP), Variance to Target %

Volume and Efficiency (4)

- * Total Units, Total Transactions, Avg Basket Size
- * Revenue per Sq Ft

Time Intelligence (6)

- * Revenue YTD, Revenue SPLY, Revenue vs SPLY %
- * Revenue Rolling 3M, Revenue MoM %, Revenue Cumulative

Ranking (5)

- * Store Rank (Company), Store Rank (Region), Region Avg Revenue
- * vs Region Avg %, Is Bottom Quartile

Dynamic (2)

- * Dynamic Chart Title (SELECTEDVALUE), Adjusted Gross Margin % (What-If)

EZ Adjusted (4)

- * EZ Adjusted Cost (SUMX), EZ Adjusted Gross Profit
- * EZ Adjusted Gross Margin %, EZ Relief Impact (ppts)

EZ Ranking (2)

- * Store Margin Rank EZ Adjusted (Region), Is Bottom Quartile EZ Adjusted